Note

It is to be noted that electrons are not the only particles with which wave nature can be demonstrated. The waves are associated with particles like neutrons and alpha particles also when they are in motion. They undergo diffraction when they are scattered by suitable crystals. Neutron diffraction studies are highly useful for investigating crystal structures.

Note

Diffraction is one of the properties of waves. Whenever waves are incident on an obstacle, they bend around the edges of the obstacle. This bending of waves is called diffraction. The amount of bending depends on the wavelength of the waves.

We have learnt in unit 6 that as the wavelength of light is very small, diffraction effects of light are very small. In order to study diffraction of light, diffraction gratings are used.

Since x-rays and de Broglie waves of electrons have wavelengths (in the order of 10^{-10}m) much shorter than that of the light wave, diffraction grating cannot be used in x-ray diffraction studies. In a crystal, the spacing between atomic planes is comparable to the wavelength of x-rays and de Broglie waves of electrons. Hence, in x-ray diffraction studies, the crystals are used which serve as three-dimensional grating.

Figure 8.18 shows the variation of intensity of the scattered electrons with the angle θ for the accelerating voltage of $54V$. For a given accelerating voltage V, the scattered wave shows a peak or maximum at an angle of 50° to the incident electron beam. This peak in intensity is attributed to the constructive interference of electrons diffracted from various atomic layers of the target material. From the known value of interplanar spacing of Nickel, the wavelength of the electron wave was experimentally calculated as 1.65Å.

The wavelength can also be calculated from de Broglie relation for V = 54 V from equation (8.18).

$$\lambda = \frac{12.27}{\sqrt{V}}\text{ Å} = \frac{12.27}{\sqrt{54}}\text{ Å}$$

$$\lambda = 1.67\text{ Å}$$

This value agrees very well with the experimentally observed wavelength of 1.65Å. Thus this experiment directly verifies de Broglie's hypothesis of the wave nature of moving particles.

8.3.5 Electron Microscope

Principle

This is the direct application of wave nature of particles. The wave nature of the electron is used in the construction of microscope called **electron microscope**.

The resolving power of a microscope is inversely proportional to the wavelength of the radiation used for illuminating the object under study. Higher magnification as well as higher resolving power can be obtained by employing the waves of shorter wavelengths.

Louis de Broglie wavelength of electron is very much less than (a few thousands less) that of the visible light being used in optical microscopes. As a result, the microscopes employing de Broglie waves of electrons have very much higher resolving power than

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